

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	30 June 2026	Maximum Marks	4 Marks
Team ID		Project Name	Strategic Product Placement Analysis System

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	User Management & Authentication	Registration through Corporate Email
		Role-based Login (Analyst, Store Manager, Admin)
		Password Recovery and Reset via Email
FR-2	Data Ingestion Module	Upload POS Sales Data (CSV format validation)
		Upload Store Floor Plan Images (PNG/JPG parsing)
FR-3	Spatial Mapping & Analysis Engine	Generate digital coordinate grid over uploaded floor plan
		Assign product sales metrics to specific grid coordinates
		Run predictive algorithm for high-traffic aisle identification
FR-4	Heatmap Visualization	Render interactive D3.js heatmap overlay on floor plan
		Toggle visibility layers (Overall Traffic vs. Specific Product)
FR-5	Reporting & Export	Generate PDF Strategy Report with placement recommendations

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
		Export raw placement coordinate data as CSV for store staff

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

NFR No.	Non-Functional Requirement	Description
NFR-1	Usability	The system must feature an intuitive UI/UX requiring less than 1 hour of training for non-technical retail staff to navigate the dashboard effectively.
NFR-2	Security	All user passwords must be hashed using bcrypt. Uploaded POS sales data must be encrypted at rest (AES-256) and transmitted securely via HTTPS (TLS 1.2+).
NFR-3	Reliability	The core application and spatial mapping engine must ensure a 99.9% uptime during standard retail business hours.
NFR-4	Performance	The system must be capable of parsing POS CSV files up to 50MB and rendering the corresponding interactive heatmap overlay within 3 seconds.
NFR-5	Availability	The web-based dashboard must be accessible 24/7 across major modern browsers (Chrome, Edge, Safari, Firefox).
NFR-6	Scalability	The system architecture must support concurrent spatial processing for up to 100 distinct retail floor plans without performance degradation.